Session 7. The cost of capital: WACC, cost of equity (CAPM), beta estimation and cost of debt

At the end of this session, you must be able to: define the cost of capital and write the WACC formula with the after-tax cost of debt; write the CAPM equation for the cost of equity and explain each input; justify the choice of the risk-free rate (zero-coupon government bond matching the cash-flow horizon); describe the three approaches to estimate the market risk premium (survey, historical, implied) with their limitations; estimate beta top-down by regression, justify the practical choices (period, frequency, index, dividends), and link beta to its determinants (cyclicality, operating leverage, financial leverage) including the levered/unlevered beta formula; and estimate the cost of debt through recent borrowing history, public ratings or synthetic ratings (interest coverage ratio).

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1. Defining the cost of capital and the WACC

> **Cost of capital:** the minimum rate of return on the company's investments that can satisfy both shareholders (the cost of equity) and debtholders (the cost of debt). It is the company's total cost of financing.

- The rate of return required by investors depends on the risk arising from the **assets-in-place** (capital employed): specifically its **systematic risk**, since unsystematic or specific risks are not remunerated.
- The cost of capital is shaped by the economic characteristics of each sector of activity: the cost structure (fixed vs variable costs); the sensitivity to the economic environment; the predictability of the activity; the results growth rates.

$WACC = k_d(1 - t_c) \cdot D/(D+E) + k_e \cdot E/(D+E)$
where k_d = expected return on debt, k_e = expected return on equity, t_c = corporate income tax rate, D = market value of debt, E = market value of equity.

- Debt enters **after tax** (interest is tax deductible) and both weights use **market values** of debt and equity.

2. The cost of equity: the CAPM

$k_e = r_f + \beta_{levered} \times MRP$
where r_f = the current risk-free rate, $\beta_{levered}$ = beta of the asset being analyzed, MRP = market risk premium.

3. The risk-free rate in theory and in practice

Risk-free asset: an asset on which the actual return equals the expected return (no variance around the expected return). Two conditions must be met:
1. **No default risk**: generally implies a security issued by the government - but not all governments can be viewed as default free.
2. **No uncertainty about reinvestment rates**: implies a **zero-coupon** security with the **same maturity as the cash flow** being analyzed.

- Proxy in the CAPM: the rate on a zero-coupon government bond matching the time horizon of the cash flow being analyzed.

The class test: in business valuation we estimate cash flows forever (or at least for very long periods). The right risk-free rate to value a company in US dollars is therefore the long-term government bond rate (the 30-year Treasury bond logic of matching horizon), NOT the 3-month T-bill: short-term bills mismatch the horizon and embed reinvestment risk. (Formally, the ideal instrument is a zero-coupon government bond matched to the cash flows' horizon - hence, among the proposed choices, the longest maturity is the best proxy.)

4. The market risk premium: definition and three estimation approaches

Market risk premium (MRP): the premium investors demand for investing in an average-risk investment, relative to the risk-free rate. As a general proposition it should be: greater than zero; increasing with the risk aversion of investors in that market; increasing with the riskiness of the "average" risk investment.

Three estimation approaches:

- **Survey approach**: survey a few (larger) investors, in practice, surveys of money managers' expectations of stock returns over the next year. Limitations: no constraints on reasonability (could produce negative premiums or premiums of 50%). Extremely volatile. Short term (even the longest surveys in this category do not go beyond one year). Scope note: the one-year cap describes the category of expectations surveys shown on the slide, not every premium survey. The Fernandez survey provided with this course measures the MRP and risk-free rate that practitioners **used** to calculate required returns (96 countries, 2024), a different object from a one-year-ahead expectation and not bounded by that cap.
- **Historical approach**: sensitive to several choices: 1. **Time period**: the MRP is time-varying; different periods produce different measures; shorter periods have higher standard errors. 2. **Choice of risk-free security**: T-bills vs T-bonds; if the term structure is upward sloping, the MRP is larger when estimated over shorter-term risk-free assets; must be consistent with the risk-free

rate chosen for the cost of capital. 3. **Arithmetic vs geometric average**: arithmetic when returns are serially uncorrelated and the estimate targets next year; geometric when returns are serially correlated (Fama and French, 1988) - it adjusts for the long-term view.

- Two further caveats even after choosing period, security and average: 1. **Noisy estimates**: even with about 80 years of history (back to 1928) and a 20% standard deviation of annual stock returns, the standard error is greater than 2% (20%/√80 = 2.26%). And the implied standard deviation in equities rose to almost 50% in the last quarter of 2008 - think of the consequences for historical premiums if such volatility persisted. 2. **Survivorship bias**: using US equity data over the twentieth century creates a sampling bias - only companies that "survived" are observed, and US markets were among the most successful of the century.
- **Implied premium**: the third approach, derived from current market prices (presented alongside the other two in the estimation process).

5. Estimating beta top-down: the regression

> **Top-down approach:** the standard procedure regresses stock returns (R_j) against market returns (R_m): $R_j = a + bR_m$, where a is the intercept and b the slope. The **slope of the regression is the beta of the stock** and measures its riskiness.

6. Beta estimation in practice: the four choices

Setting up the estimation:

- **Estimation period (length)**: services use 2 to 5 years. Longer periods give more data, but firms change; shorter periods can be affected by significant firm-specific events (example: ITT for 1995-1997).
- **Return interval (frequency)**: daily, weekly, monthly. Shorter intervals yield more observations but suffer from more noise. The slide says this noise is created by stocks not trading and "biases all betas towards one". Treat that direction as the slide's own shortcut, not a demonstrated universal law: a later slide describes only a bias in beta estimates caused by the lack of liquidity of certain stocks, without fixing its direction.
- **Returns**: include dividends. Return = $(\text{Price}_{\text{End}} - \text{Price}_{\text{Beginning}} + \text{Dividends}_{\text{Period}}) / \text{Price}_{\text{Beginning}}$; include dividends only in the ex-dividend month.
- **Market index**: estimate returns (inclusive of dividends) on the index for each interval; given internationally diversified investors, it is increasingly common to compute beta against international indices, such as the Morgan Stanley Capital Index.

Summary of the major problems:

- **Historical returns**: longer period = more observations but possible bias from structural changes in the firm's/industry's risk profile - the main reason to support a shorter period in the 2-5 year range.
- **Return type**: both stock and market returns should include capital gains AND dividends; some sources use price changes only.
- **Frequency**: daily returns increase observations but expose the estimation to a significant bias from lack of liquidity of certain stocks.
- **The market index**: international indices are increasingly used.

7. The determinants of beta

Three determinants, with their implications:

- **Nature of the product or service** (drives the beta of the firm): the more discretionary the product or service, the higher the beta. Implications: cyclical companies > noncyclical; luxury goods > basic goods; high-priced goods/services > low-priced; growth firms have higher betas.
- **Operating leverage** (fixed costs as % of total costs): the greater the proportion of fixed costs, the higher the beta. Implications: firms with high infrastructure needs and rigid cost structures > firms with flexible cost structures; smaller firms > larger firms; young firms > mature firms.
- **Financial leverage**: other things equal, the greater the proportion of capital raised from debt, the higher the equity beta. Implications: highly levered firms have higher betas; equity β (levered) = unlevered β × adjustment.

8. Levered and unlevered beta

$\beta_L = \beta_u(1 + (1 - t_c) \times D/E)$
where β_L = levered or equity beta, β_u = unlevered beta, t_c = corporate marginal tax rate, D = market value of debt, E = market value of equity.

- The beta of equity can be written as a function of the unlevered beta and the debt-equity ratio: leverage amplifies the equity beta, mitigated by the tax shield $(1 - t_c)$.

9. Beta for non-traded assets: the bottom-up approach

- Conventional regression approaches do not work for assets that are not traded.
- Solution: **use comparable firms** (see the Heinz example in class) - estimate betas of comparable traded firms, unlever them, and relever at the target's capital structure.

10. The cost of debt: determinants

> **Cost of debt (R_d):** the rate at which a company can **currently** borrow, given a certain debt maturity. It reflects default risk and the level of interest rates in the market.

The cost of debt is driven by:

1. **Risk-free rate**: the basic component - as the risk-free rate increases, the cost of debt increases.
2. **Default risk of the company**: as default risk (and the associated default spread) increases, the cost of debt increases. It is somehow related to country risk (e.g. cost of debt for Italian companies in 2016), but be careful: not all companies have equal exposure to country risk.
3. **Tax shield on debt**: interest on debt is tax deductible, so the after-tax cost of debt is a function of the marginal corporate tax rate:

After-tax Cost of Debt = Pre-tax Cost of Debt × $(1 - t_c)$

11. Three approaches to estimate the cost of debt

- **Recent borrowing history**: look at the yield to maturity on a liquid straight bond recently outstanding from the firm. Limitation: very few firms have long-term straight bonds that are liquid and widely traded.
- **Public rating of the company**: use the firm's rating and estimate a default spread based on it. More robust, but different bonds from the same firm can have different ratings - use a **median rating** for the firm.
- **Synthetic rating**: when there are no ratings or multiple ratings, estimate a synthetic rating and base the cost of debt on it. Synthetic rating as a function of the interest coverage ratio: ICR = EBIT / Interest Expenses. Other models exist (e.g. Altman's Z-score), beyond the scope of this first part.
- Useful data source: Damodaran's ratings page (pages.stern.nyu.edu/~adamodar) for default spreads by rating.

12. Common mistakes and exam traps

- Using book values of D and E in the WACC: the formula uses **market values**.
- Forgetting the $(1 - t_c)$ on debt in the WACC - or applying it to equity as well: only debt is tax-deductible.
- Picking the 3-month T-bill as risk-free rate for valuing a company: mismatch of horizon and reinvestment risk; match a zero-coupon government bond to the cash-flow horizon (long-term bond in practice).
- Mixing up the determinants of beta: product cyclicality and operating leverage drive the **firm** (unlevered) beta; financial leverage drives the amplification from unlevered to levered beta.
- Omitting dividends from the returns used in the beta regression (both stock and index).
- Believing daily returns are always better for the regression: more observations, but noise from non-trading that biases beta estimates (the direction "towards one" is the slide's shortcut, not an established universal).
- Using the coupon rate or the rate at issuance as the cost of debt: it is the rate at which the company can borrow **currently**, for the given maturity.
- Averaging several bond ratings of the same firm arbitrarily: use the **median** rating.
- Forgetting the survival-bias and noise caveats when quoting a historical MRP (standard error above 2% even with 80 years of data).

13. Must know absolutely

Must know:
• Cost of capital = minimum return satisfying shareholders and debtholders. Driven by systematic risk of assets-in-place. Shaped by cost structure, sensitivity to the environment, predictability, growth rates.
• WACC = $k_d(1-t_c)D/(D+E) + k_eE/(D+E)$, market values, after-tax debt.
• CAPM: $k_e = r_f + \beta_L \times MRP$. Risk-free = zero-coupon government bond matched to the horizon. MRP > 0, increasing in risk aversion and average risk.
• MRP estimation: survey (volatile, short-term, unbounded), historical (sensitive to period, risk-free choice, arithmetic vs geometric average. Noisy. Survivorship bias), implied premium.
• Beta top-down = slope of $R_j = a + bR_m$. Choices: 2-5 years, return interval (daily = more observations but non-trading noise. The claimed "towards one" direction is the slide's shortcut, not established), include dividends, international index increasingly used.
• Beta determinants: discretionary products, high operating leverage, high financial leverage all raise beta.
• $\beta_L = \beta_u(1 + (1-t_c)D/E)$. Non-traded assets → bottom-up with comparables.
• Cost of debt = current borrowing rate. drivers: risk-free rate, default risk (country risk with unequal exposure), tax shield. estimation via recent borrowing history, public rating (median), synthetic rating (ICR = EBIT / interest expenses).

Watch Out

Watch out: in the WACC, the tax adjustment $(1 - t_c)$ applies **only to debt**. Applying it to equity, or forgetting it on debt, are the two most frequent computational errors on this topic.

Watch out: not all companies have equal exposure to country risk when the default spread is estimated - a multinational incorporated in a risky country may deserve a smaller spread than a purely domestic firm.